

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I **THAMARAISELVI S (192511042)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

THAMARAISELVI S

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1) CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM:

(a) Propositional Logic Representation:

Let:

F = Fever

C = Cough

R = Rash

B = Breathlessness

FL = Possible Flu

M = Possible Measles

P = Possible Pneumonia

Rules:

1. $F \text{ AND } C \rightarrow FL$
2. $F \text{ AND } R \rightarrow M$
3. $C \text{ AND } B \rightarrow P$

Facts:

F = True

C = True

B = True

(b) Forward Chaining:

Initial facts:

Fever = True

Cough = True

Breathlessness = True

Step 1:

$F \text{ AND } C \rightarrow FL$

Since Fever and Cough are true:

Possible Flu = True

Step 2:

$F \text{ AND } R \rightarrow M$

Rash is not given, so this rule cannot be applied.

Step 3:

$C \text{ AND } B \rightarrow P$

Since Cough and Breathlessness are true:

Possible Pneumonia = True

Final diagnoses:

1. Possible Flu
2. Possible Pneumonia

Possible Measles cannot be derived because Rash is not given.

(c) Backward Chaining:

Goal:

Possible Pneumonia

Rule:

$C \text{ AND } B \rightarrow P$

To prove Pneumonia, we need:

Cough = True

Breathlessness = True

Both facts are already available.

Therefore:

Possible Pneumonia = True

Goal Tree:Pneumonia

Cough AND Breathlessness

True True

Therefore, Patient A has Possible Pneumonia.

2)CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT:

(a) Unification and MGU:

Rule 1:

$\text{Vehicle}(x) \text{ AND } \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Given:

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

Match x with Ambulance:

$x = \text{Ambulance}$

Substitution:

$\theta = \{x/\text{Ambulance}\}$

Therefore:

$\text{ClearPath}(\text{Ambulance}) = \text{True}$

MGU:

$\theta = \{x/\text{Ambulance}\}$

(b) Resolution Proof for Proceed(CarA):

Relevant rules:

Rule 1:

$\text{Vehicle}(x) \text{ AND } \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

CNF:

$\text{NOT Vehicle}(x) \text{ OR NOT EmergencyType}(x) \text{ OR ClearPath}(x)$

Rule 2:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x)$

CNF:

$\text{NOT ClearPath}(x) \text{ OR GreenSignal}(x)$

Rule 3:

Vehicle(x) AND Behind(x,y) AND GreenSignal(y) -> Proceed(x)

CNF:

NOT Vehicle(x) OR NOT Behind(x,y) OR NOT GreenSignal(y) OR Proceed(x)

Given facts:

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA,Ambulance)

Step 1:

Vehicle(Ambulance) + EmergencyType(Ambulance)

Therefore:

ClearPath(Ambulance)

Step 2:

ClearPath(Ambulance)

Therefore:

GreenSignal(Ambulance)

Step 3:

Vehicle(CarA)

Behind(CarA,Ambulance)

GreenSignal(Ambulance)

Therefore:

Proceed(CarA)

Hence:

Proceed(CarA) = True

(c) Two Emergency Vehicles:

The FOL knowledge base can represent multiple emergency vehicles because the rules use variables.

For a second emergency vehicle, we need facts such as:

Vehicle(Ambulance2)

EmergencyType(Ambulance2)

Then:

Vehicle(Ambulance2) AND EmergencyType(Ambulance2)

-> ClearPath(Ambulance2)

Additional priority rules may be required if two emergency vehicles need different priorities.

Therefore, the existing structure is reusable, but additional facts are needed for each emergency vehicle.

3) CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM:**(a) CNF Conversion:**

Let:

S = SoilDry

I = IrrigationNeeded

W = CropWheat
D = ApplyDripMethod
R = CropAtRisk

P1:

S $\rightarrow$ I

CNF:

NOT S OR I

P2:

I AND W $\rightarrow$ D

Remove implication:

NOT(I AND W) OR D

Using De Morgan's law:

NOT I OR NOT W OR D

CNF:

NOT I OR NOT W OR D

P3:

NOT D AND W $\rightarrow$ R

Remove implication:

NOT(NOT D AND W) OR R

Using De Morgan's law:

D OR NOT W OR R

CNF:

D OR NOT W OR R

Facts:

S = True

W = True

(b) Resolution Refutation:

Goal:

D = ApplyDripMethod

Negated goal:

NOT D

Clauses:

C1: NOT S OR I

C2: NOT I OR NOT W OR D

C3: D OR NOT W OR R

C4: S

C5: W

C6: NOT D

Resolution Step 1:

$C1 + C4$

NOT S OR I

S

Therefore:

I

Resolution Step 2:

$C2 + I + W$

NOT I OR NOT W OR D

Therefore:

D

Resolution Step 3:

$D + \text{NOT } D$

Therefore:

Empty Clause ($\square$)

Hence:

ApplyDripMethod is proved.

(c) Removing SoilDry:**If SoilDry is removed:**

IrrigationNeeded cannot be derived from Rule P1.

Therefore:

ApplyDripMethod cannot be derived.

However, CropAtRisk also cannot be concluded.

This is because Rule P3 requires:

NOT ApplyDripMethod AND CropWheat

Removing ApplyDripMethod does not mean NOT ApplyDripMethod is true

Therefore:

CropAtRisk does not follow.

4) CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT:**(a) Belief State:**

Given percepts:

1. Stench at [1,2]
2. Breeze at [1,1]
3. Glitter at [2,2]

Rule 1:

$\text{Stench}[1,2] \rightarrow \text{WumpusAdjacent}[1,2]$

Therefore:

$\text{WumpusAdjacent}[1,2] = \text{True}$

Rule 2:

Breeze[1,1] $\rightarrow$ PitAdjacent[1,1]
Therefore:
PitAdjacent[1,1] = True

Rule 3:

Glitter[x,y] $\rightarrow$ Gold[x,y]
Therefore:
Gold[2,2] = True

Derived facts:

1. Wumpus is adjacent to [1,2].
2. Pit is adjacent to [1,1].
3. Gold is at [2,2].

(b) Forward Chaining:**Step 1:**

Stench[1,2]
Therefore:
WumpusAdjacent[1,2]

Step 2:

Breeze[1,1]
Therefore:
PitAdjacent[1,1]

Step 3:

Glitter[2,2]

Therefore:
Gold[2,2]

Thus:

Possible Wumpus location = a cell adjacent to [1,2]

Possible Pit location = a cell adjacent to [1,1]

The exact locations cannot be determined from the given rules.

(c) Safety of the Path:**Path:**

[1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]
The knowledge base tells us:
PitAdjacent[1,1] = True
WumpusAdjacent[1,2] = True

Safety rule:

NOT Wumpus[x,y] AND NOT Pit[x,y] $\rightarrow$ Safe[x,y]

The KB does not prove that [1,2] is free from Wumpus and Pit.

Therefore, the complete path cannot be confirmed as safe.

Conclusion:

The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is NOT logically proven safe.

Gold at $[2,2]$ is confirmed, but additional information is needed to establish the safety of the path.